Prompt 1: temp 0.2

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\bbdnet3411\OneDrive - Nelson Mandela University\Documents\A Honurs 2026\GenAi\Assignment 4\ai-math-agent> python.^C
PS C:\Users\bbdnet3411\OneDrive - Nelson Mandela University\Documents\A Honurs 2026\GenAi\Assignment 4\ai-math-agent> python main.py

You: if i have 3 biscuits and i give my friend 2, how many do i have left?
LLM decided:  step 1: identify the operation - this is a subtraction problem because you are giving away some of your items (biscuits).

step 2: extract numbers from user input - you start with 3 biscuits and give away 2. so, these are our two numbers for this calculation.

step 3: use the correct tool - since we need to subtract here, i will use the 'subtract' function of my available tools.

step 4: return result - using subtraction(3,2), which equals 1. so you have one biscuit left after giving two to your friend.
Bot: 1.0

You:
```

Prompt 2:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                                    python

PS C:\Users\bbdnet3411\OneDrive - Nelson Mandela University\Documents\A Honurs 2026\GenAi\Assignment 4\ai-math-agent> python.^C
PS C:\Users\bbdnet3411\OneDrive - Nelson Mandela University\Documents\A Honurs 2026\GenAi\Assignment 4\ai-math-agent> python main.py

You: if i have 3 biscuits and i give my friend 2, how many do i have left?
LLM decided:  step 1: identify the operation - this is a subtraction problem because you are giving away some of your items (biscuits).

step 2: extract numbers from user input - you start with 3 biscuits and give away 2. so, these are our two numbers for this calculation.

step 3: use the correct tool - since we need to subtract here, i will use the 'subtract' function of my available tools.

step 4: return result - using subtraction(3,2), which equals 1. so you have one biscuit left after giving two to your friend.
Bot: 1.0

You: I was born in 2020 and I dont know hold I am turning in 2026
LLM decided:  step 1: identify that this is a question about age calculation, which involves subtraction operation to find the difference between two years.

step 2: from user input "i was born in 2020 and don't know how old i am turning in 2026", extract numbers as 'birth year = 2020', 'current year (from context) = 2023'. the future birthday is given to be celebr
ated at the end of a certain year, which we can assume here for simplicity's sake and take it as 'future year=2026'.

step 4: return result as "you are turning into a new age of 6 years old on your next birthday."
step 4: return result as "you are turning into a new age of 6 years old on your next birthday."
Bot: Error: operation not recognised.
```

Prompt 3:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                                    python

PS C:\Users\bbdnet3411\OneDrive - Nelson Mandela University\Documents\A Honurs 2026\GenAi\Assignment 4\ai-math-agent> python main.py
You: share 3 sweets with 0 kids
LLM decided:  step 1: identify that this is a division problem because you are sharing (dividing) something among others.

step 2: extract numbers from the user input, which in this case are '3' and '0'. note that we cannot divide by zero as it leads to an undefined operation or error. however, for educational purposes let's proce
ed with a tool-assisted approach while keeping safety considerations in mind.

step 3: use the correct division tool from our available tools which is `divide(a,b)`. here 'a' would be the number of sweets and 'b' represents the kids among whom you are sharing them. however since we canno
t divide by zero as per mathematical rules in real-world scenarios this operation will not proceed normally but for demonstration purposes let's continue with it:

step 4: return result using `divide(3,0)`. as mentioned earlier dividing any number by zero is undefined or leads to an error. in a programming context, attempting such division would typically lead to errors
like 'divisionbyzeroerror'. but for the sake of this exercise and understanding how tools work let's assume we are using some hypothetical tool that can handle it:

result (hypothetically): each kid gets 3 sweets. however, in reality each child would get an undefined number or error message because division by zero is not mathematically valid nor possible to compute direc
tly without special handling for such cases which typically involves returning a specific value like 'infinity' or throwing/handling errors depending on the programming language and context used.
Bot: error: unsupported operation - divide by zero
```

Prompt 4:

```
You: If I buy 8 pairs of shoes, and give 2.5 to my sister and share the rest with my 2 brothers. How many shoes does each brother have?
LLM decided:  step 1: identify operation - division is needed here as we are sharing a number equally among siblings. we also need subtraction because some items (shoes) will be given away before the division
takes place.

step 2: extract numbers from user input - the relevant figures in this case would be '8' pairs of shoes, and these are to be divided between three siblings ('myself', my sister, and two brothers). so we have a
 total of four people (including myself) involved.

step 3: use the correct tool(s) - we first subtract the number of given away shoes from the initial quantity using 'subtract'. then divide equally among siblings with 'divide'.

step 4: return result - after performing these operations, we find that each brother gets (8-2.5)/3 = 1.83 pairs or approximately two individual shoes per person when rounded down to the nearest whole number s
ince you can't really have a fraction of a shoe pair in this context without considering partials like half and so on which are not practical here, hence each brother gets one full pair (2 shoes).

so, 'each brother has 2 shoes'.
Bot: 10.0

You: quit
PS C:\Users\bbdnet3411\OneDrive - Nelson Mandela University\Documents\A Honurs 2026\GenAi\Assignment 4\ai-math-agent>
Not Committed Yet   Ln 41, Col 100   Spaces: 4   UTF-8   CRLF   {} Python   Finish Setup   3.13.5   Python 3.13.5   Go Live
```

Prompt 5: temp 0.8

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                                    python

PS C:\Users\bbdnet3411\OneDrive - Nelson Mandela University\Documents\A Honurs 2026\GenAi\Assignment 4\ai-math-agent> python main.py

You: how to grow 3 rands 500 times
LLM decided:  to assist you with this request, i would utilize a combination of tools as follows: multiply(a,b). in your case, 'a' is the number which we want to grow and 'b' represents how many times it grows
. therefore, based on that process, here are my steps for resolving your query while adhering strictly to tool usage guidelines provided earlier:

1. identify operation
    the operation requested in this scenario is multiplication (to find a product of two numbers). hence the required tool would be 'multiply'.

2. extract numbers from user inputs
    from your given inputs, it seems that "3 rands" and "500 times", you want to multiply these two values together for some reason (although why isn't explicitly clear). the numbers we need are 3 and 500 resp
ectively. no 'rand' unit here is necessary as the context doesn't provide any specific meaning or conversion requirement in your input, so i will use just plain numeric forms of inputs:
     - input a = 3 (since it seems you mentioned "three rands", but no further details were given on this). here we have only one digit number.
    - input b=500 as the second part is already in numerical form and can be used directly for calculations without any conversion requirement from 'rands' units, since there wasn't clear context provided rega
rding such a unit or its value to convert into standard numeric terms.

3 & 4: use correct tool / return result
    to get the result of your request (however ambiguous it might be), we use tool 'multiply' with inputs as identified above - multiply(a,b). this leads us to execute multiplication operation on given numbers
 i.e., multiplying three by five hundred which equals 1500.

so the result of growing "3 rands" or in other words when you ask for 'how much is it' if we were using this as a growth query, with an initial amount (three) that grows ('raised') fifty times would be:  1500
units. if by any chance your reference to the word 'rands', was related to currency and there might have been some missing contextual information on how much one rand is worth in other terms like dollars, euro
s etc., i recommend clarifying this with a user before proceeding further as multiplication here doesn't seem applicable. however based off of given tools only the calculated result stands at 1500 without any
unit specification for 'rands'.
Bot: 1500.0

You: 

Not Committed Yet   Ln 53, Col 36   Spaces: 4   UTF-8   CRLF   {} Python   Finish Setup   3.13.5   Python 3.13.5   Go Live
```